IYOBOR PROGRESSIVE UNION

Contribution & Loan App

System Documentation

Item	Detail
Version	1.0 — system as built to 12 October 2026
Status	Built and tested; trial site live; 2026–2027 records loaded (March–October)
Live address	https://ipu-app.netlify.app (trial)
Source code	github.com/festus-oliha/ipu-app (private)
Prepared for	Iyobor Progressive Union, 198, Old Sapele Road, Benin City, Edo State
System administrator	Festus Ozee (Admin)
Developer	Application built by JVeeSoft-IT
Date	12 October 2026

Related documents: Requirements Specification and System Design (approved 11/10/2026).

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1. Introduction

1.1 Purpose

This document describes the Iyobor Progressive Union (IPU) Contribution and Loan App as it has been built: what it does, who uses it, the technology it runs on, how the money rules are implemented, how data is stored, and how the system is configured, deployed and operated. It is the reference for the Admin and for any developer who maintains the app in future.

1.2 Background

Until 2026 the union kept its records in an Excel workbook with one sheet per month (March to February). The workbook worked, but formulas were copied by hand each year and some errors had crept in: unpaid interest was dropped when a member took a top-up loan, a credit for overpaid interest was wiped when a loan was cleared, and one month's formulas pointed at the wrong rows. The app replaces the workbook with a single, shared system that applies the union's rules the same way every time, keeps a permanent record of every change, and lets members see their own position on their phones.

1.3 Scope of this version

Version 1.0 covers the full money cycle of the union:

- members, logins and roles; recording contributions, fees, interest and repayments; receipts uploaded by members;
- loans with limit checks, guarantor consent, approval by the President then the Treasurer, and payout;
- monthly interest, month close with bank reconciliation and sign-off; investments, other income and expenses;
- member exit, year-end settlement with dividend, debtors, default recovery and legal referral;
- SMS and email notifications, scheduled reminders, statements, reports and Excel exports;
- the import of the 2026–2027 workbook (March to October 2026).

Items still to be built are listed in section 13.

2. System overview

The app is a secure website that works on any phone or computer browser. Members sign in with their phone number and a password. What each person sees depends on their role. All money is kept in one ledger (a list of entries that is never edited or deleted), and every figure on every screen — savings, loan balances, interest, bank balance, statements — is worked out from that ledger.

2.1 Users and roles

Role	Who	What they can do
Member	Every member	See own savings, loan, interest due and payment history; upload receipts; request a loan; accept or decline being a guarantor; print own statement; request exit.
Admin	Festus Ozee	Add and edit members, give roles, reset passwords; open and close financial years; set the dividend; reopen a closed month; send test SMS and run reminders; download full records.
President	Madam F. U. Idahosa	Approve or reject loans (first approval) and exceptions; approve investments; sign closed months; decide exits; declare defaults; refer for legal action; record payments.
Treasurer	Mrs Felicia Imade	Second loan approval and payout; close each month against the bank statement; confirm receipts; investments, income and expenses; pay exits and year-end payouts; recover defaulted loans from savings.
Secretary	Mr O. Egbenehikhuere	Add and edit members, reset passwords; record payments; sign closed months; record contacts with debtors.
Auditor	(when appointed)	Read-only view of all office pages, reports and full records.

> **Separation of duties.** No officer can record a payment, approve, pay out or check a receipt on their own account. The President and Treasurer approvals of a loan must be by different people, and the two month-end signatures must be by different people.

2.2 Main screens

Area	Screens
Member	Home (savings, loan, interest due, receipts, guarantees, payment history); Upload receipt; Request a loan; Loan details; My statement; Leave the union
Office	Monthly sheet (with month close and sign-off); Record payment; Receipts to check; Loans; Members; Debtors
Office → More	Bank and investments; Members leaving (exits); Year end; Messages (SMS/email outbox); Reports
Public	Sign-in; Contact us

3. Technology stack

The app uses widely supported, current technologies chosen so that the union is not tied to one developer: the code is standard TypeScript, the database is ordinary PostgreSQL, and every service can be replaced.

Layer	Technology	Version	Role in the app
Web framework	Next.js (App Router, Server Actions, Turbopack)	16.4.0	Pages, forms and server logic in one app; pages are rendered on the server so phones download little.
User interface	React	19.3.0	Interactive forms (live loan-limit check, receipt photo shrinking, approvals).
Styling	Tailwind CSS	4	Brand colours (forest green, gold, ivory) and phone-first layout.
Fonts	Marcellus, Public Sans (bundled via @fontsource)	5.3	Headings and body text; bundled so the site never depends on Google Fonts.
Language	TypeScript	5 (ES2022)	Type-checked code; money handled as whole kobo (BigInt).
Database	PostgreSQL on Supabase	Supabase managed	All records; transaction pooler for the app, session pooler for migrations.
Data access	Prisma ORM with @prisma/adapter-pg	7.10.0	Schema, migrations and type-safe queries.
File storage	Supabase Storage (private bucket ipu-files)	—	Receipt photos and PDFs; never public, streamed only after a permission check.
Sessions	iron-session (encrypted, HTTP-only cookie)	9.0	Sign-in; signed out after 30 minutes without activity.
Passwords	Argon2 (@node-rs/argon2)	2.2	Password hashing; 5 wrong attempts lock the login for 15 minutes.
Validation	Zod	4	Checking form input on the server.
SMS	Termii HTTP API (adapter in src/lib/messaging.ts)	—	Payment confirmations and reminders; provider can be swapped.
Email	Nodemailer over SMTP (e.g. Gmail)	7	Guarantor reports and the monthly backup.
Spreadsheets	ExcelJS	4.4	Excel exports (month sheets, full records) and reading the old workbook.
Hosting	Netlify (Next.js runtime + scheduled function)	—	Live website and the hourly reminder job.
Source control	Git and GitHub	—	Version history; each push can deploy to Netlify.
Testing	Vitest; Playwright (browser tests)	5; 1.x	36 automated rule tests; end-to-end browser checks.
Code quality	ESLint, TypeScript compiler	9; 5	Checks run before every commit.

3.1 Architecture

Members and officers use the web app in their browser. The app runs on Netlify and talks to the Supabase database and file storage. Outgoing SMS and email are written to an outbox and sent straight after each action, and again by an hourly scheduled function, so a slow provider never blocks an officer. Code changes are pushed to GitHub, from which Netlify builds and publishes the site.

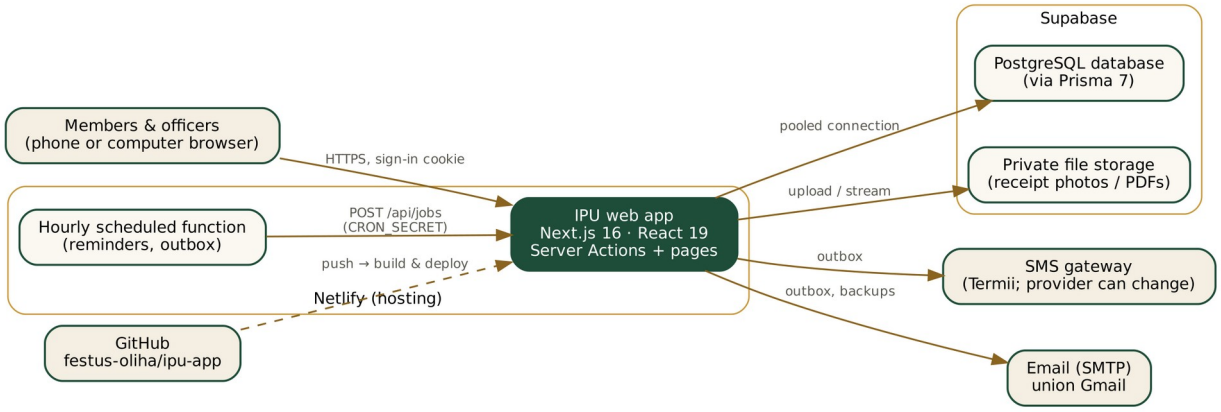


Figure 1. System architecture

4. Business rules as implemented

All money rules live in one tested module (src/lib/rules.ts and src/lib/position.ts). The same code drives the screens, the month close, statements, exits and the year end. Amounts are stored in kobo (₦1 = 100 kobo) as whole numbers, so there are no rounding errors.

4.1 Interest

- **Rate:** ₦25 per ₦1,000 of loan balance per month (2.5%).
- **When charged:** when the Treasurer closes a month, on the loan balance at the end of that month. It is due by the 15th of the next month (SMS sent at close).
- **Unpaid interest** becomes arrears and is added to the next month's interest. No penalty and no interest on interest.
- **Overpaid interest** is kept as a credit and refunded at year end (or at exit).
- **February's interest** is charged when February is closed, before the year-end settlement.

Each member has one loan account: **loan balance** = loans given − loans repaid − loan taken from savings; **interest owed** = interest charged − interest paid − interest taken from savings + credit refunded. A positive interest owed is arrears; a negative one is a credit.

4.2 Loans

- **Limit:** usual monthly contribution × 12. Above the limit a loan is an **exception** and needs approval by the President or Treasurer; the guarantor is informed.
- **Window:** loans are given from March to December only; never in January or February.
- **Guarantor:** one per loan; a member may guarantee at most two open loans. The guarantor must accept in the app (or the Secretary records a signed paper form).
- **Approval:** guarantor → President → Treasurer; then the Treasurer records the payout (date and bank reference).
- **Repayment:** any amount at any time; the loan is marked repaid when nothing is owed.
- **Loans from the 2026–2027 workbook** were agreed verbally; they are recorded with "verbal guarantee (before the app)". New loans always need a guarantor.

4.3 Contributions

- Members may pay more or less than their usual amount in any month; the usual amount is only a guide and the basis of the loan limit.
- **Payment ahead (batch):** one payment can cover several months (e.g. ₦60,000 = 4 × ₦15,000). It fills the next months not yet paid for and counts for each of them (for the dividend's four-month rule) while the money is counted in the bank in the month it arrived.
- **Savings** = contributions paid in the year − anything taken from savings to clear a debt (own or as guarantor).

4.4 Month close and sign-off

1. The Treasurer closes a month once it has ended and all uploaded receipts for it are checked.
2. The app compares its bank balance (opening balance + money in − money out) with the GT Bank statement; any difference needs a written explanation.
3. Interest is charged for every borrower and SMS reminders are queued.
4. The President and the Secretary sign; with both signatures the month is locked. Only the Admin can reopen the latest closed month, with a reason; reopening reverses that month's interest charges.

4.5 Year end

- **Dividend:** one flat percentage set by the union each year, paid on savings, only to members who contributed in at least 4 months.
- **Payout** = savings + dividend + interest credit − interest owed − loan balance.
- **Debtors** (who owe more than they have): savings and dividend clear interest first, then the loan; the rest is either carried into the new year or must be paid by a set date (with reminders).
- **Payouts** are recorded by the Treasurer as bank transfers in the new year.

4.6 Exit, default and recovery

- **Exit:** a member may leave at any time; they get back this year’s savings less anything owed; no dividend. A guarantor of an open loan must be replaced first.
- **Default:** the President or Treasurer declares a loan in default; borrower and guarantor are told at once. After 7 days the Treasurer may recover the debt from the borrower’s savings first, then the guarantor’s.
- **Legal referral:** the Admin or President can refer a debtor for legal action and track the case.

5. Data model

The database has the following main tables. Every table has an internal identifier; personal data is visible only to officers.

Table	Holds
User, UserRole	Logins (phone, Argon2 password hash, lock-out counters) and roles.
Member	Member number (IPU-0001…), name, phone (optional until known), bank details, usual monthly contribution, status (active, suspended, exited).
FinancialYear, Period	Years (March–February), their 12 months, month status (open, ready, signed, locked), bank statement figure and signatures.
LedgerEntry	Every movement of money: date, month, member, type, amount, bank reference, note, who entered it, and months covered for payments made ahead. Never edited or deleted; a correction is a reversal entry.
Loan, LoanApproval, LoanMonth	Loan requests and their workflow, each approval or rejection, and the frozen month-by-month schedule.
Receipt	Uploaded receipt (file key in storage, amounts, status, who checked it).
Investment	MMF, treasury bills, fixed deposits: proposal, approvals, placement and return.
Exit, Settlement	Exit requests and payouts; year-end settlement per member (dividend, deductions, payout, debtor plan).
ContactLog, LegalReferral	Calls, visits and promises to pay; legal cases.
Notification	The SMS and email outbox: status, attempts, cost and last error.
Setting	Union settings by date (e.g. the opening bank balance of each year).
AuditLog	Every change: who, when, what, old and new values, reason.

5.1 Ledger entry types

Type	Bank effect	Meaning
CONTRIBUTION	In	Monthly contribution (may cover several months)
MEMBERSHIP_FEE	In	Joining or membership fee
INTEREST_PAID	In	Interest paid by a borrower
LOAN_REPAID	In	Loan repayment
OTHER_INCOME	In	Other money received
INVESTMENT_RETURNED	In	Investment principal and profit back in the bank
LOAN_GIVEN	Out	Loan paid out
INVESTMENT_PLACED	Out	Money placed in an investment
EXPENSE	Out	Bank charges, SMS bundles and other costs
PAYOUT	Out	Exit or year-end payout
INTEREST_CHARGED	None	Monthly interest charged at month close
DEDUCTION	None	Taken from a member’s savings to clear a debt
LOAN_DEDUCTED / INTEREST_DEDUCTED	None	Loan or interest cleared from savings
CREDIT_REFUNDED	None	Overpaid interest given back
REVERSAL	None	Cancels a wrong entry (the original stays visible, struck

Type	Bank effect	Meaning
		through)

Schema changes are applied with database migrations kept in prisma/migrations: init (11/10/2026), phase4 (deductions, exit and settlement details), phase5 (outbox subject and error), phase6 (optional phone, verbal guarantor, months covered).

6. Notifications and scheduled jobs

Every message is first written to the outbox (Notification table) and sent immediately after the action that created it. An hourly scheduled function on Netlify calls the app at /api/jobs with a shared secret; the app then runs whatever is due today (Lagos time) and retries anything still waiting. Each job runs only once per day or period.

When	What	Who is told
Each payment recorded or receipt confirmed	Amount received, savings this year, loan balance	Member (SMS)
Loan steps	Guarantor request, approvals, exception, payout	Borrower, guarantor (SMS)
Month close	Interest due by the 15th and loan balance	Each borrower (SMS)
10th of the month	Reminder of interest due	Borrowers who still owe (SMS)
16th of the month	Unpaid interest has become arrears	Borrower and guarantor (SMS, email)
1st and 15th, December–February	Year-end warning of what will be taken from savings	Members owing (SMS)
Last day of May, Aug, Nov, Feb	Quarterly loan report	Each guarantor (email, or SMS)
Daily	Investment matures in 7 days	Treasurer (SMS, email)
1st and 15th	Pay-up date reminder	Debtors on a pay-up plan (SMS)
1 and 20 February	Open next year before 1 March	Admin (SMS); also a notice on the Admin home page
1st of the month	Full records backup (Excel)	Admin (email)
Year close	Payout or amount still owed	Every member (SMS)

> **Current setting:** SMS is in test mode (SMS_PROVIDER = log) and email is not configured, so messages are recorded in the outbox but not delivered. They start going out once the SMS provider and the union Gmail are set up (section 8).

7. Security, privacy and audit

- **Sign-in:** phone number and password; passwords hashed with Argon2; at least 8 characters; changed at first sign-in; 5 wrong attempts lock the account for 15 minutes; sessions end after 30 minutes without activity.
- **Permissions:** every page and every action checks the user's role on the server, even if a button is hidden. Members can only ever load their own records and files.
- **No self-dealing:** officers cannot record, approve, pay out, check receipts or recover savings on their own accounts.
- **Ledger integrity:** money entries are never edited or deleted. Corrections are reversals with a reason; closed months need the Admin to reopen them.
- **Change log:** every change to money or member data is written to the audit log with who, when, old and new values and reason.
- **Files:** receipts are stored privately in Supabase Storage and streamed only to the owner and officers; there are no public file links. The file type is checked from its contents, not its name.
- **Secrets:** database passwords, keys and the session secret are kept only in the .env file on the Admin's computer and in Netlify's environment settings; they are never committed to GitHub.
- **Privacy:** phone numbers and bank details are visible only to officers, in line with the Nigeria Data Protection Act 2023. Exports with personal data are limited to the Admin and Auditor.
- **Backups:** Supabase keeps daily database backups; the app also emails the Admin a full Excel copy on the 1st of each month (once email is set up).

8. Configuration

Settings are environment variables: in the .env file on the Admin's computer, and under Project configuration → Environment variables in Netlify. A Netlify change takes effect at the next deploy.

Variable	Purpose	Needed
DATABASE_URL	Supabase transaction pooler (port 6543) used by the app	Yes
DIRECT_URL	Supabase session pooler (port 5432) used for migrations and imports	Yes
SESSION_SECRET	32+ random characters protecting sign-in cookies	Yes
SUPABASE_URL	Supabase project address (https://xxxx.supabase.co)	Live site
SUPABASE_SERVICE_ROLE_KEY	Supabase secret key (sb_secret_…) for receipt storage	Live site
CRON_SECRET	32+ random characters shared with the hourly function	Live site
SMS_PROVIDER	termii, log (test mode) or empty (messages wait)	For SMS
TERMII_API_KEY, TERMII_SENDER_ID	Termii account key and registered sender name (e.g. IPU)	For SMS
TERMII_BASE_URL, TERMII_CHANNEL	Optional; channel defaults to dnd (reaches Do-Not-Disturb numbers)	Optional
SMS_UNIT_COST	Naira per SMS page, to show SMS spend	Optional
SMTP_HOST, SMTP_PORT, SMTP_USER, SMTP_PASS	Email account (Gmail: smtp.gmail.com, 465, app password)	For email
EMAIL_FROM, BACKUP_EMAIL	Sender name and backup address	Optional
ADMIN_PHONE, ADMIN_NAME, ADMIN_TEMP_PASSWORD	First Admin account (seed only; not needed on Netlify)	Setup only

9. Deployment and operations

9.1 Running on the Admin's computer

5. Install Git, Node.js 22 and VS Code; clone github.com/festus-oliha/ipu-app.
6. In the VS Code terminal: npm install
7. Create .env from .env.example and fill in the values (section 8).
8. Apply database changes: npm run db:deploy (first time also npm run db:seed).
9. Start the app: npm run dev and open http://localhost:3000

The computer and the live site use the same Supabase database, so changes made on either appear on both.

9.2 Publishing the live site (Netlify)

- The Netlify project ipu-app is connected to the GitHub repository and is **public** (required so members and the hourly function can reach it; data is protected by sign-in).
- Builds are currently **stopped** to save credits. To publish the latest code: Project configuration → Build & deploy → Build settings → Builds: Active, then Deploys → Trigger deploy.
- **Rule:** whenever the database changes (npm run db:deploy or an import), publish the latest code too, otherwise the live site and the database disagree.

9.3 Everyday commands

Command	Purpose
npm run dev	Run the app on the computer
npm run db:deploy	Apply new database migrations (after a git pull that adds one)
npm test	Run the money-rule tests
npm run lint / npm run typecheck	Check the code
npm run import:workbook <file.xlsx>	Dry run of a workbook import (nothing saved)
npm run import:commit <file.xlsx>	Save a workbook import
npm run db:studio	Look inside the database

9.4 Running the union year

When	Who	Task
Any day	Treasurer, Secretary, President	Record payments; check uploaded receipts
After each month ends	Treasurer	Close the month against the GT Bank statement
After close	President and Secretary	Sign the month (it then locks)
From 1 February	Admin	Open the next financial year (reminder shown)
Early March	Treasurer; Admin	Close February; set the dividend and close the year
March onwards	Treasurer	Record each year-end payout transfer

10. Data migration: 2026–2027 workbook

The records for March to October 2026 were loaded from the union workbook by a dedicated import tool (scripts/import-workbook.ts). The tool reads each member's monthly payment, loan given, interest paid and loan returned; it lets the app work out the interest with its own rules, closes March to September, leaves October open, and compares the result with the workbook month by month. It always runs as a dry run first.

10.1 Result

- 18 members (16 from the workbook plus the officers), with logins for every member who has a phone number.
- All contributions, loans and repayments match the workbook; the bank balance agrees every month (₦972,250 at October).
- Mrs Felicia Imade's ₦20,000 monthly payment is split: ₦10,000 for her and ₦10,000 for Madam F. U. Idahosa, who pays through her.
- Mrs Vero Ekeoba's two ₦60,000 payments are recorded as paid ahead (April–July and August–November at ₦15,000).
- Precious Abraham (Mama Tope) and Enotieomwan Daughter2 have no phone number yet and therefore no login.

10.2 Corrections to the workbook

Member	Workbook	App (correct)	Cause
Mr Austin Aikhiome Ohiole	Interest owed ₦35,250 (September)	₦43,250	The ₦8,000 June interest was dropped when he took a top-up loan in July.
Mrs Enotieomwan Grace	Interest ₦0 after repayment	Credit ₦2,500 (refunded at year end)	April's formulas pointed at the wrong rows, and the credit was wiped when the loan was cleared.

A corrected copy of the workbook (formulas fixed) was supplied for running Excel alongside the app; it now agrees with the app in every closed month.

11. Testing and quality

- **Rule tests (Vitest):** 36 automated tests of interest, arrears, credits, loan limit, dividend, year-end payout, exit, settlement and default recovery, using real cases from the 2025 and 2026 workbooks.
- **Browser tests (Playwright):** complete journeys on a phone-sized screen: adding members, recording payments, the loan approval chain, month close and sign-off, receipt upload and checking, investments, exits, year end, reminders, debtors and recovery, reports and imports.
- **Checks before every release:** TypeScript type check, ESLint and a full production build.
- **Data check:** the import compares every month’s figures with the workbook before anything is saved.

12. Source code structure

Path	Contents
prisma/schema.prisma, prisma/migrations/	Database tables and the history of changes
prisma/seed.ts	First Admin account, the 2026–2027 year and its months, union settings
src/lib/rules.ts, position.ts	The union’s money rules (tested)
src/lib/ledger.ts, loans.ts, payments.ts	Ledger totals, loan accounts, recording payments
src/lib/monthClose.ts, yearEnd.ts, settlement.ts	Month close, year end, exits and deductions
src/lib/notify.ts, messaging.ts, jobs.ts	Outbox, SMS and email sending, scheduled jobs
src/lib/storage.ts	Private receipt storage (Supabase or local folder)
src/lib/statement.ts, sheet.ts, exportWorkbook.ts	Statements, the monthly sheet, Excel exports
src/lib/auth.ts, session.ts	Sign-in, roles and sessions
src/app/(app)/…	Signed-in screens (member and office pages, server actions)
src/app/login, contact, api/jobs, files, exports	Sign-in, Contact us, scheduled-job endpoint, file and export downloads
src/components/	Shared pieces: seal, footer, letterhead, forms
netlify/functions/hourly.mts	Netlify scheduled function (hourly)
scripts/import-workbook.ts	Workbook import tool

13. Known limitations and next steps

13.1 Still to build

- Officer dashboard (key figures on one screen).
- Member profile page (update contact details, change password).
- Change-log screen for the Admin and Auditor (the log is already recorded).
- Forgot password by SMS code (today the Admin or Secretary resets passwords).
- Settings page for rates and dates (today in code).
- Membership certificates, ID cards and loan letters with signatures and reference numbers (designs ready in the brand kit).

13.2 Before full go-live

- Choose the SMS provider, fund it and register the sender name “IPU”; set the SMS variables in Netlify.
- Set up the union Gmail app password for email.
- Publish the latest code on Netlify (builds are currently stopped).
- Give members their temporary passwords; add phone numbers for members without one.
- Run the app alongside the corrected workbook for one full month, then retire the workbook.
- Optional: a proper web address (e.g. ipu.com.ng) and moving the Netlify project to a union-owned account.

14. Version history

Date	Change
11/10/2026	Phase 1: project set-up, database design, money rules, sign-in
11/10/2026	Phase 2: members, recording payments, monthly sheet; Supabase database
11/10/2026	Phase 3: loans, guarantor, approvals, interest, month close and sign-off
11/10/2026	Phase 4: receipts, bank and investments, exits, year end
11/10/2026	Phase 5: SMS and email, scheduled reminders, debtors, reports and exports; trial site on Netlify
11/10/2026	Workbook import; batch contributions; members without phone; verbal guarantors
11/10/2026	Footer, Contact us page, year-change reminders
12/10/2026	This document (version 1.0)